26 January 1976
SUPERSEDING
MIL-V-22854A
2 March 1965

MILITARY SPECIFICATION

VALVES, ANGLE, SHUT-OFF, PACKED, RECEIVER, REFRIGERANT-12

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of manually operated shut-off valve for use with Refrigerant-12 (see 6.1).

1.2 Classification. Valves shall be of the following sizes as specified (see 6.2):

Size	National Pipe Thread (NPT) Inlet (Inch)	Flare and Solder Outlet (Inch)
1	1/4	1/4 flare
2	3/8	3/8 flare
3	3/8	1/2 flare
4	1/2	1/2 flare
5	1/2	5/8 flare
6	3/4	7/8 solder
7	1/2	3/8 flare

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATION

FEDERAL

MIL-V-3 - Valves, Fittings and Flanges (Except for Systems Indicated Herein); Packaging of

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MS 17243 - Valves, Angle, Shut-Off, Packed, Receiver, Refrigerant-12

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Standard product. Valves delivered under this specification shall be the manufacturer's standard commercial catalog product, except for any modifications and additional requirements specified herein.

3.2 Materials. Materials shall conform to the applicable specifications, standards, and requirements specified herein. Materials not definitely specified shall be of the quality normally used by the manufacturer in fabricating his valves.

3.3 Design and construction. The design and construction of the valves shall be as specified herein and as shown on MS 17243. Sizes shall be as indicated in 1.2 and on MS 17243. The valves shall be of the angle, packed, back-seating, or nonbackseating type with a gasketed seal cap. A wrench grip shall be provided on each valve. Threads and the seating surface of the flare connection shall be protected with plastic caps or by other means to prevent damage during handling.

3.4 Finish. Finish shall be the manufacturer's standard commercial finish.

3.5 Marking. The manufacturer's standard identification and other data standard with the manufacturer shall be placed on all valves. Unmarked valves will be accepted where the manufacturer's standard practice does not require marking.

3.6 Workmanship. All parts of the finished valves shall be clean and free from pits, sprues, scale, flux or extraneous material. External surfaces shall be free from burrs and sharp edges.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Sampling for inspection and acceptance. Sampling for inspection and acceptance shall be performed in accordance with the provisions set forth in MIL-STD-105, except where otherwise indicated.

4.2.1 Component and material inspection. In accordance with 4.1 above, the supplier is responsible for insuring that components and materials used are manufactured and inspected in accordance with the requirements of referenced subsidiary specifications and standards to the extent specified or, if none, in accordance with this specification.

4.2.2 End product inspection. The inspection lot shall consist of all valves of one size offered for inspection at one time. The sample unit for this examination shall be one valve.

4.2.2.1 Visual examination. Examination of the end item shall be in accordance with table I. The inspection level shall be II with an acceptable quality level (AQL) of 2.5 for major defects and 6.5 for total defects, expressed in terms of defects per hundred units.

TABLE I. Classification of defects

Examine	Defect	Classification Major	Minor
Design, construction and workmanship, general (applicable to all components and assemblies)	Wrench grip not provided	X	
	Sharp burr, sliver, splinter, pits, scale, or sharp edges	X	

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TABLE I. Classification of defects (cont'd)

Examine	Defect	Classification Major	Classification Minor
Threads and seating surface	Not protected from damage (plastic wrap or other means of protecting thread not provided)		X
	Damaged	X	
Marking (as applicable)	Missing, not legible		X

4.2.2.2 Dimensional examination. Inspection shall be made of the valves for compliance with dimensions and threads specified in MS 17243. Any deviation from specified requirements shall constitute a defect. The inspection level shall be S-2 with an AQL of 4.0 defects, expressed in terms of defects per hundred units.

4.2.2.3 End product testing. The supplier shall perform the test in accordance with 4.3.1. The inspection level shall be S-2 with an AQL of 4.0 defects expressed in terms of defects per hundred units.

4.2.3 Inspection of preparation for delivery. An inspection shall be made in accordance with MIL-V-3 to determine compliance with section 5 requirements.

4.3 Test.

4.3.1 Angle valve leak test. The angle valve, with the seat open and the discharge end blocked, shall be submerged in water and a pressure of 350 pound-per-square-inch gage pressure of air shall be applied to the intake to check for body leakage. The pressure shall be released and the seat closed and the discharge end unblocked. The valve shall be submerged in water again and the same pressure applied. Any leakage through the body or by the seat shall be cause for failure of this test.

5. PREPARATION FOR DELIVERY

* 5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.2).

* 5.1.1 Levels A and C. Valves shall be preserved and packaged in accordance with the applicable requirements of MIL-V-3.

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* 5.2 Packing. Packing shall be level A, B or C as specified (see 6.2).

* 5.2.1 Levels A, B and C. Valves, preserved and packaged as specified in 5.1, shall be packed in accordance with the applicable requirements of MIL-V-3.

* 5.3 Marking. In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with the applicable requirements of MIL-V-3.

6. NOTES

6.1 Intended use. Valves covered in this specification are intended for use in a refrigerating system as a means of shutting-off the system's charge of refrigerant and containing it in the liquid receiver or condenser receiver.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Size of valves required (see 1.2).
(c) Selection of applicable levels of preservation, packaging and packing (see 5.1).

6.3 The outside margins of this document have been marked to indicate where changes from the previous issue have been made. An asterisk (*) indicates a substantive change in the paragraph text. This has been done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and suppliers are cautioned to evaluate the requirements of this document based on the entire content as written irrespective of the marginal notations and relationship to the last previous issue.

Custodians:

Army - GL
Navy - YD
Air Force - 82

Preparing activity:

Army - GL

Project No. 4130-0154

Review activity:

DSA - GS

User activities:

Navy - MC, CG

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE MIL--V-22854B - VALVES, ANGLE, SHUT-OFF, PACKED, RECEIVER, REFRIGERANT-12

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

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